

ECE 456s: Selected Topics in Signal Processing

Course Project Description

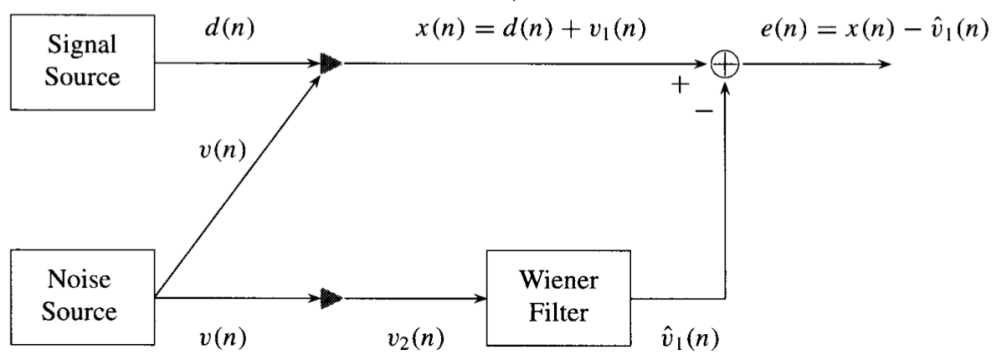
Spring 2026

Introduction:

- This project is composed of two separate tasks:
 1. Noise Cancellation Using an FIR Wiener Filter
 2. Adaptive Linear Prediction Using the LMS Algorithm
- For both tasks you can use Matlab or GNU-Octave or Python to achieve the requirements.

Task 1: Noise Cancellation Using an FIR Wiener Filter (40%)

Consider the noise cancellation system shown below.



- Suppose that the desired signal $d(n)$ is a sinusoidal signal

$$d(n) = \sin(n\omega_0 + \phi)$$

with $\omega_0 = 0.05\pi$.

- (10%) Let the noise sequences $v_1(n)$ and $v_2(n)$ be first-order autoregressive processes that are generated by the first-order difference equations

$$v_1(n) = 0.8v_1(n-1) + g(n)$$

$$v_2(n) = -0.6v_2(n-1) + g(n)$$

where $g(n)$ is zero-mean, unit variance white noise that is uncorrelated with $d(n)$.

- (20%) Write a Matlab code in order to design an FIR Wiener filter that can be used for noise cancellation. Let the filter order take the values 6 and 12.
- (10%) Generate plots of the desired signal contaminated with the noise, and of the “cleaned” signal after noise cancellation for filter orders 6 and 12.

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Task 2: Adaptive Linear Prediction Using the LMS Algorithm (60%)

- (10%) Let $x(n)$ be a second-order autoregressive process that is generated according to the difference equation

$$x(n) = 1.2728x(n-1) - 0.81x(n-2) + v(n)$$

where $v(n)$ is unit variance white noise.

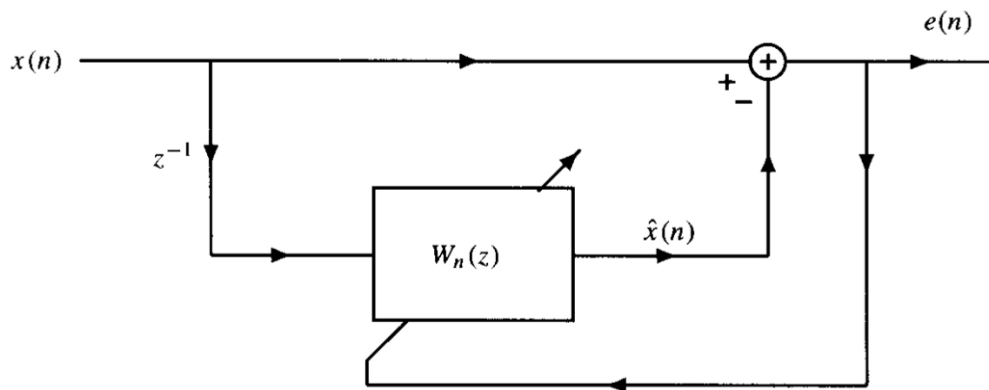
- It can be shown that the optimum causal linear predictor for $x(n)$ is

$$\hat{x}(n) = 1.2728x(n-1) - 0.81x(n-2)$$

- It is required to use the LMS algorithm to build an adaptive linear predictor of the form

$$\hat{x}(n) = w_n(1)x(n-1) + w_n(2)x(n-2)$$

as shown in the figure below.



- (30%) Write a Matlab code to build an adaptive linear predictor using the LMS algorithm. Let the initial values for $w_n(1)$ and $w_n(2)$ be zeros.
- (10%) Generate plots of the convergence of the filter coefficients and the prediction error with iterations for step size of 0.02 and 0.04.
- (10%) Use Matlab to compute the power spectral density of the process $x(n)$, and hence compute its autocorrelation function, then determine the eigenvalues of the autocorrelation matrix from which you can get an upper bound on the step size for the LMS algorithm.

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ECE Program

Electronics & Communications Engineering Program

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General Instructions:

1. This is a group project. Each group should have two/three students.
 2. Please prepare one compressed file that includes the following items:
 - (a) Your Matlab or Python codes (*.m file or *.py file)
 - (b) A report (pdf files Only) that includes the detailed steps of each task and relevant plots of the waveforms.
 3. Project should be submitted on LMS before 11 : 59 PM on May 15th 2026.
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Good Luck